

Exercise No. 3

STATISTICAL TESTS AND PSEUDORANDOM NUMBERS GENERATORS

Adam Cherek, s207250

3.1

Plain text:

Perfect pancakes are easier to make than you think. This pancake recipe produces thick, fluffy, and all-around delicious pancakes with just a few ingredients that are probably already in your kitchen (and it's so much better than the boxed stuff). You likely already have everything you need to make this pancake recipe.

$\alpha = 0,05$	Frequency test	Serial test	Poker test	Runs test
	max = stats pass?	max = stats pass?	max = stats pass?	max = stats pass?
Plain text				
Caesar cipher				
Hill cipher				
DES				
AES				

Conclusions:

- Plaintext & Caesar Cipher: Standard English text is highly non-random (ASCII bit representations contain heavy biases, e.g., leading zeros). The Caesar cipher is a monoalphabetic substitution cipher; it merely shifts characters but preserves the exact statistical frequency of the plaintext bitstream. Therefore, both fail all statistical tests completely.
- Hill Cipher: As a polygraphic substitution cipher based on linear algebra, it hides single-letter frequencies better than Caesar, but it still exhibits strong linear patterns and poor diffusion. It fails to mimic a truly random sequence, thus failing most or all tests.
- DES & AES: Both are modern block ciphers designed with strict Confusion and Diffusion properties (the Avalanche effect). Even highly structured plaintext is mapped to ciphertext that is statistically indistinguishable from random noise. Therefore, both DES and AES easily pass all pseudorandomness tests.

3.2

$\alpha = 0,05$	Frequency test max = stats pass?	Serial test max = stats pass?	Poker test max = stats pass?	Runs test max = stats pass?
LCG1				
LCG2				
LFSR1				
LFSR2				
BBS1				
BBS2				

Conclusions:

- Parameters dictate quality: The structure of a PRNG algorithm alone does not guarantee randomness. LCG and BBS can either perfectly pass or completely fail statistical testing strictly based on the parameters chosen.
- LCG constraints: LCG1 (Park-Miller) provides good statistical randomness, but LCG2 produces a trivially predictable alternating sequence because its modulus $m=2$ is far too small.
- LFSR polynomials: LFSR1 passes because a primitive polynomial ensures a maximal period. LFSR2 fails because non-primitive polynomials result in shortened periods and repeating bit patterns that are quickly caught by the Poker and Runs tests.
- BBS state space: The BBS algorithm is provably secure, but only if the modulus $M = p \times q$ is large enough to prevent short sequence loops. BBS1 fails due to a tiny modulus (77), while BBS2 behaves securely due to its larger prime parameters.